



S2: Advanced Statistical Methods

Yuchen Mao (ym429)

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Abstract

High-content phenotypic screening, and especially the Cell Painting assay, has revolutionized early-stage drug discovery by enabling rapid morphological profiling of cells under thousands of perturbations. However, fluorescence-based imaging is costly, time-consuming, and prone to artefacts, limiting its scalability. Computational prediction of immunofluorescence (IF) from label-free bright-field (BF) microscopy offers a promising alternative, but existing convolutional and adversarial models suffer from blurred outputs or unstable training, while diffusion models incur high inference costs.

Here, we reproduce and extend the CellResDM framework—a residual-shifted diffusion model that efficiently predicts five-channel Cell Painting IF from a single BF image in as few as 15 steps, over $60\times$ faster than standard diffusion, with strong fidelity to both pixel-level and single-cell morphological features. We further introduce Consistency Distilled CellResDM, integrating consistency distillation to reduce inference to just two steps, surpassing all baselines in accuracy and efficiency on the cpg0000-jump-pilot dataset. These advances move label-free, molecularly-resolved cell profiling closer to practical routine use in large-scale drug discovery.

1 Introduction

High-content, phenotypic high-throughput screening (HTS) is now foundational in early-stage drug discovery, enabling systematic, large-scale profiling of how thousands of chemical and genetic perturbations alter cell behaviour. Among these, the *Cell Painting* assay [1, 2] has emerged as a gold standard for unbiased morphological profiling. By multiplexed staining of key subcellular structures—nucleus, endoplasmic reticulum (ER), actin/Golgi/plasma membrane (AGP), mitochondria, and RNA—Cell Painting generates high-dimensional morphological “fingerprints” for each cell. These fingerprints are rich in information, making them invaluable for single-cell profiling, compound clustering, and hit triage in both academic and industrial screens.

Despite its broad utility, fluorescence imaging presents **significant practical limitations**: (i) *Cost and availability*—specialised dyes (e.g., TOTO-3 for DNA) can cost over \$500 per 384-well plate and are prone to shortages; (ii) *Throughput*—capturing five high-resolution channels per plate adds hours to each experiment; and (iii) *Phototoxicity and artefacts*—fluorescent labels may perturb cell physiology and introduce unwanted variability, complicating downstream analyses.

These limitations have spurred interest in **label-free alternatives**, foremost among them *bright-field (BF) microscopy*. BF imaging rapidly captures structural information from unstained, live cells

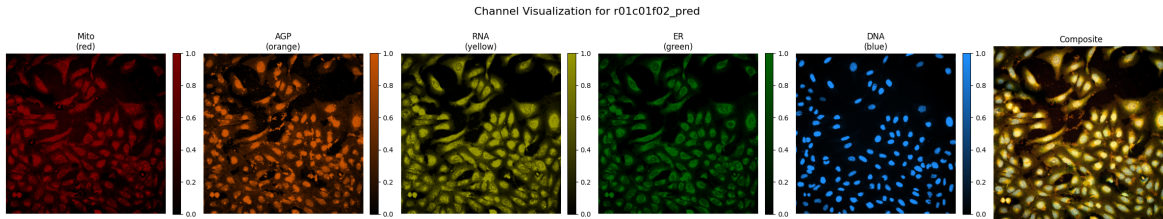


Figure 1: **Five-channel Cell Painting field-of-view.** Each panel shows a single fluorescence channel (mitochondria, AGP, RNA, ER, DNA), with the rightmost panel merging all channels into a pseudo-colour composite constructed for visualization reused throughout this manuscript [2].

using transmitted light, providing inexpensive, dye-free, and non-destructive snapshots across entire screening plates. However, BF inherently lacks the molecular specificity of fluorescence: fine protein localization and many functional compartments are invisible, limiting its use for deep mechanistic profiling.

This motivates a key computational challenge: *can we recover the full molecular richness of Cell Painting fluorescence channels directly from simple BF images?* Recent machine learning work has begun to address this gap: early convolutional neural networks (CNNs) [3] achieved promising reconstructions of nuclear and cytoskeletal markers; GANs [8, 9, 15] improved image sharpness, but often hallucinated features or suffered from instability; and state-of-the-art diffusion models [4–6, 17] have further improved realism, albeit at considerable computational cost. Critically, many prior studies have not thoroughly assessed whether the generated images preserve *single-cell morphological statistics*—the very features essential for robust screening, mechanistic interpretation, and quantitative analysis.

In this work, we rigorously reproduce and validate **CellResDM**, an efficient residual-shifted diffusion model for five-channel Cell Painting synthesis from BF images, and extend it with **Consistency Distilled CellResDM**, which employs consistency distillation to accelerate inference and further improve fidelity. On the cpg0000-jump-pilot benchmark, our models achieve state-of-the-art pixel accuracy and single-cell feature correlation, reducing inference steps from 15 to just 2 while outperforming all tested baselines. Together, these advances bring label-free, molecular-level cell profiling closer to practical, scalable use in drug discovery—cutting costs, eliminating dye artefacts, and crucially, preserving the biological information needed for downstream analytics.

2 Methods

2.1 Overview

We focus on the CellResDM framework, an efficient residual-shifted diffusion model for translating label-free bright-field (BF) images into multi-channel immunofluorescence (IF) Cell Painting images. Standard diffusion models generate images by denoising from pure noise, essentially constructing an output “from scratch.” This process requires many inference steps and is unnecessarily complex when there is already a strong structural correspondence between the input and target. By contrast, CellResDM exploits the close relationship between BF and IF by learning to remove both noise and only the residual difference between the two modalities at each step of the diffusion process. This approach reframes generation as predicting the minimal changes needed to transform a BF image into its corresponding IF, dramatically reducing task complexity and thus enabling high-fidelity synthesis in as few as 15 steps.

Building on this foundation, we incorporate consistency distillation—yielding Consistency Distilled CellResDM—which further compresses the denoising trajectory to just two inference steps, with no sacrifice in accuracy or feature preservation. We rigorously compare both models against baseline methods and provide comprehensive evaluation using both image-level and biologically meaningful single-cell metrics.

The remainder of this section details the mathematical formulation of CellResDM (forward and reverse processes), the integration of consistency distillation, the noise scheduling strategy, and the experimental protocols used throughout this work.

2.2 CellResDM: Residual-Shifted Diffusion Model

Given a target immunofluorescence (IF) image x_0 and a source bright-field (BF) image y_0 , we define the residual:

$$e_0 = y_0 - x_0$$

which encodes the pixel-wise difference between BF and IF images.

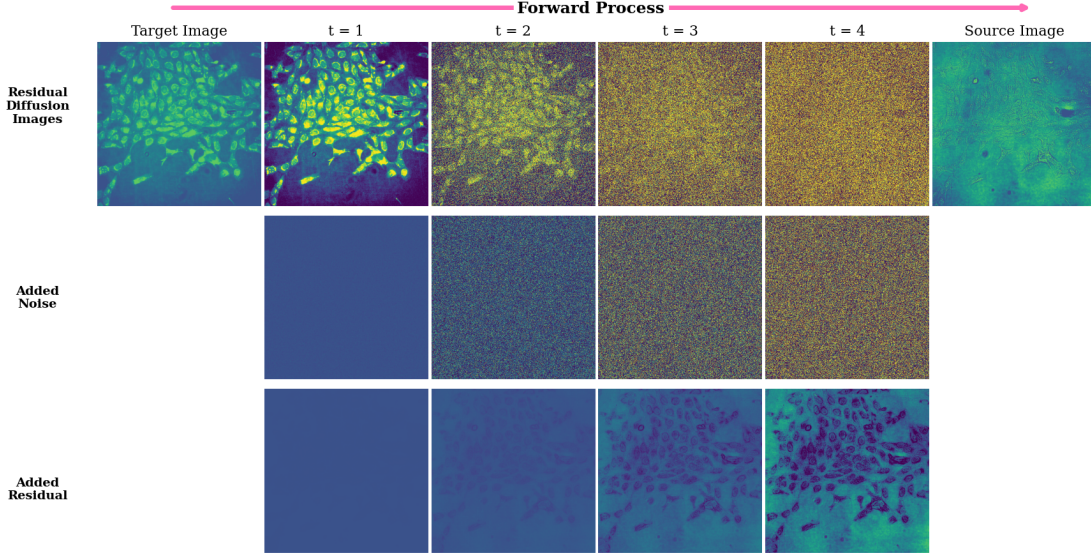


Figure 2: Visualization of the residual diffusion process between target (mitochondrial) and source (BF) image over a 4-step diffusion. Top: residual diffusion at each step. Middle: noise added at each step. Bottom: residuals.

Forward Process. Let $\{\eta_t\}_{t=0}^T$ be a monotonically increasing sequence of coefficients, with $\eta_0 = 0$, $\eta_T \approx 1$. The forward process, at each step, shifts the current image by a fraction of the residual and adds noise:

$$q(x_t|x_{t-1}, y_0) = \mathcal{N}(x_t; x_{t-1} + \alpha_t e_0, \kappa^2 \alpha_t I) \quad (1)$$

where $\alpha_t = \eta_t - \eta_{t-1}$, and κ sets the relative noise scale.

This Markovian definition fully explains our forward process, but it's inefficient for directly sampling x_t during training. By induction (see Appendix A.1), the process can be re-parametrized as:

$$q(x_t|x_0, y_0) = \mathcal{N}(x_t; x_0 + \eta_t e_0, \kappa^2 \eta_t I) \quad (2)$$

That is, for any t , we can write:

$$x_t = x_0 + \eta_t e_0 + \kappa \sqrt{\eta_t} \epsilon$$

with $\epsilon \sim \mathcal{N}(0, I)$, which is easy to sample from.

Reverse Process. Given a fully noised sample x_T , our goal is to recover the target image x_0 conditioned on the clean source y_0 . This is achieved by learning a reverse Markov chain, which

iteratively de-noises x_t to x_{t-1} , ultimately reconstructing x_0 . The marginal reverse generative process can be written as:

$$p(x_0 | y_0) = \int p(x_T | y_0) \prod_{t=1}^T p_\theta(x_{t-1} | x_t, y_0) dx_{1:T} \quad (3)$$

where the prior at the final time-step is a Gaussian centred on the source image:

$$p(x_T | y_0) \approx \mathcal{N}(x_T; y_0, \kappa^2 I),$$

that is, the initial input for de-noising is a fully corrupted version of y_0 .

The reverse transitions themselves are modelled as learned Gaussians:

$$p_\theta(x_{t-1} | x_t, y_0) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, y_0, t), \Sigma_\theta(x_t, y_0, t)), \quad (4)$$

where μ_θ and Σ_θ are predicted by a neural network. Unlike the forward process, these reverse transitions do not follow directly from the diffusion construction and must be learned.

By leveraging Bayes' theorem and the Markov property of the forward process (see Appendix A.2), the true reverse posterior can be computed in closed form as:

$$q(x_{t-1} | x_t, x_0, y_0) = \mathcal{N}(x_{t-1}; \mu_t, \Sigma_t) \quad (5)$$

$$\mu_t = \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} x_0 \quad (6)$$

$$\Sigma_t = \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I \quad (7)$$

Under this construction, only x_0 is unknown during inference, so we replace x_0 in the mean with a neural network prediction $f_\theta(x_t, y_0, t)$:

$$p_\theta(x_{t-1} | x_t, y_0) = \mathcal{N}\left(x_{t-1}; \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} f_\theta(x_t, y_0, t), \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I\right) \quad (8)$$

Training Loss. The model is trained by minimizing the evidence lower bound (ELBO), which decomposes into a sum of KL divergences between the true and learned reverse transitions at each step:

$$\min_{\theta} \sum_{t=1}^T D_{\text{KL}}(q(x_{t-1} | x_t, x_0, y_0) \| p_\theta(x_{t-1} | x_t, y_0))$$

Because both conditionals are Gaussians with the same covariance, the KL divergence reduces to a weighted mean squared error (see Appendix A.3 for the full derivation):

$$D_{\text{KL}}(q \| p_\theta) = \frac{\alpha_t}{2\kappa^2 \eta_{t-1} \eta_t} \|x_0 - f_\theta(x_t, y_0, t)\|_2^2 \quad (9)$$

However, the coefficient $\frac{\alpha_t}{2\kappa^2 \eta_{t-1} \eta_t}$ becomes extremely large for small t , leading to training instability. Therefore, following prior work, we omit the coefficient and optimize an unweighted MSE loss:

$$\mathcal{L}_{\text{CellResDM}}(\theta) = \mathbb{E}_{x_0, y_0, \epsilon, t} [\|x_0 - f_\theta(x_t, y_0, t)\|_2^2] \quad (10)$$

where $x_t = x_0 + \eta_t \epsilon_0 + \kappa \sqrt{\eta_t} \epsilon$. For enhanced quality, we combine MSE with the learned perceptual image patch similarity (LPIPS) loss [18]. The LPIPS loss measures the perceptual similarity between two images by comparing their activations in a deep neural network (typically a pretrained VGG

network). Unlike pixel-wise losses, LPIPS is sensitive to human perceptual judgments of structure and texture, thus encouraging the network to produce images that are not only accurate at the pixel level but also visually realistic and structurally faithful. Formally, for two images x and $\hat{x}$,

$$\mathcal{L}_{\text{LPIPS}}(x, \hat{x}) = \sum_l w_l \|\phi_l(x) - \phi_l(\hat{x})\|_2^2, \quad (11)$$

where ϕ_l denotes the feature map from layer l of a pretrained network, and w_l are learned weights reflecting perceptual importance. Therefore, our final loss can be described as:

$$\mathcal{L}_{\text{CellResDM}}(\theta) = \mathbb{E}_{x_0, y_0, \epsilon, t} [\|x_0 - f_\theta(x_t, y_0, t)\|_2^2 + \lambda \mathcal{L}_{\text{LPIPS}}(x_0, f_\theta(x_t, y_0, t))] \quad (12)$$

where λ balances pixel wise and perceptual fidelity.

2.3 Noise Scheduling

Having introduced both the forward and reverse processes, we now detail the key hyperparameters that control the addition of noise and residual in CellResDM: the sequence of noise coefficients $\{\eta_t\}_{t=0}^T$. These coefficients are fundamental, as they determine both the magnitude and schedule of residual injection and stochastic perturbation (the noise added) at each step of the diffusion process (see Eq. 2).

Recall that, for any t , the forward process constructs noised samples as:

$$x_t = x_0 + \eta_t e_0 + \kappa \sqrt{\eta_t} \epsilon$$

where $e_0 = y_0 - x_0$ is the fixed residual, $\epsilon \sim \mathcal{N}(0, I)$, and κ is a global noise scale. The sequence $\{\eta_t\}$ smoothly interpolates from “no residual, no noise” ($t = 0$) to “full residual, maximal noise” ($t = T$).

Boundary Conditions and Design Principles. For the scheduling to be effective and numerically stable, it must satisfy several desiderata:

- **Start near the target:** At $t = 1$, $\eta_1 \rightarrow 0$, so $x_1 \approx x_0 + \text{minuscule noise}$, ensuring the first denoising step truly reflects the data distribution’s score [14].
- **End near the source:** At $t = T$, $\eta_T \rightarrow 1$, so $x_T \approx y_0 + \text{maximal noise}$, meaning the initial state for reverse diffusion is nearly pure noise centered on the BF image.
- **Avoid degeneracies:** η_1 and η_T cannot be exactly 0 or 1. If $\eta_1 = 0$, no noise is injected, resulting in a Dirac delta and loss of generalization. If $\eta_T = 1$, the last forward step collapses, destabilizing training and inference. Additionally, $\alpha_T = \eta_T - \eta_{T-1}$ becomes very small, “killing” the model update in the last reverse step.

For practical and numerical reasons, we adopt:

$$\eta_1 = \min \left(\left(\frac{0.04}{\kappa} \right)^2, 0.001 \right), \quad \eta_T = 0.99$$

Both are chosen to be very close to the theoretical limits, but always non-degenerate.

Geometric Noise Scheduling. Defining just the endpoints is insufficient; the full sequence $\{\eta_t\}_{t=1}^T$ crucially affects learning. Following [17], we use a non-uniform geometric progression for the *square*

roots of the noise coefficients:

$$\sqrt{\eta_t} = \sqrt{\eta_1} \cdot b_0^{\beta_t}, \quad t = 2, \dots, T-1 \quad (13)$$

$$\beta_t = \left(\frac{t-1}{T-1} \right)^p \cdot (T-1), \quad b_0 = \exp \left[\frac{1}{2(T-1)} \log \frac{\eta_T}{\eta_1} \right] \quad (14)$$

where p is a shape parameter (default $p = 0.3$), controlling how aggressively the schedule increases. This schedule starts with almost no noise for early steps, ramping up steeply towards the end.

Numerical Example. For our main experiments, we use $T = 15$, $\kappa = 2.0$, $p = 0.3$, and $\eta_T = 0.99$. These values were chosen for stability and best empirical trade-off between fidelity and diversity.

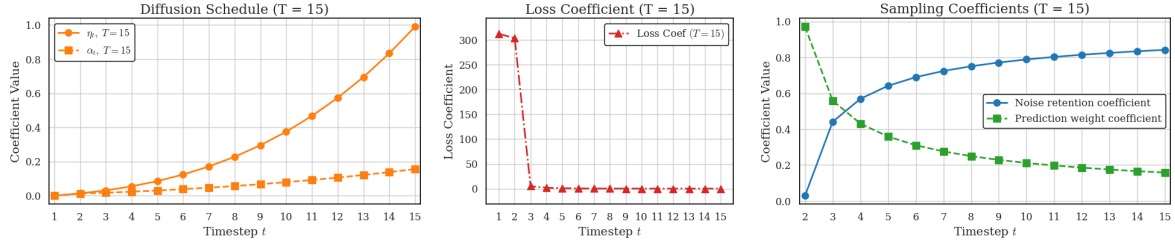


Figure 3: **Noise scheduling in CellResDM.** (Left) Training-time values of $\sqrt{\eta_t}$, showing how noise increases non-linearly over time. (Middle) Theoretical loss coefficient at each step, rapidly decaying from high weight (early, low noise) to almost zero (late, high noise). (Right) Inference-time sampling coefficients: noise retention (η_{t-1}/η_t) and prediction weight (α_t/η_t), which modulate the exploration-exploitation balance at each denoising step.

Intuition: Exploration–Exploitation Trade-off in Sampling and Training. The non-uniform geometric schedule for $\{\eta_t\}$ is central to both the inference (sampling) dynamics and the denoising task during training.

During inference (sampling): Two key coefficients orchestrate the transition from noise to prediction at each reverse step:

$$\frac{\eta_{t-1}}{\eta_t} \quad (\text{noise retention}), \quad \frac{\alpha_t}{\eta_t} \quad (\text{prediction weight}).$$

- **Early timesteps:** $\frac{\eta_{t-1}}{\eta_t} \approx 1$, so the process preserves most of the noised image, allowing exploration of possible solution space. The network’s correction ($\frac{\alpha_t}{\eta_t}$) is minimal.
- **Late timesteps:** $\frac{\alpha_t}{\eta_t}$ increases, giving more weight to the network’s prediction, which sharpens and refines the output (exploitation). Simultaneously, $\frac{\eta_{t-1}}{\eta_t}$ drops, phasing out the noise.

This smooth transition enables the model to sample diversely at first, then focus on generating high-fidelity details as inference progresses.

During training: The schedule for η_t directly governs the amount of residual e_0 and the noise $\kappa\sqrt{\eta_t}$ added to x_0 at each timestep in the forward process (Equation 2):

$$x_t = x_0 + \eta_t e_0 + \kappa\sqrt{\eta_t} \epsilon.$$

- **Early timesteps:** η_t is close to zero, so only a tiny amount of residual and noise is added. The model must learn to correct very subtle perturbations—recovering fine details and structure.

-
- **Late timesteps:** As η_t approaches one, x_t is dominated by both the full residual and strong noise, resulting in a heavily corrupted input. Here, the model thus require learning robust, coarse reconstructions from extreme noise

This progressive increase in difficulty ensures the model can handle both nearly clean and highly corrupted images, promoting robust denoising across the full trajectory.

Impact on the Loss Coefficient. The noise scheduling not only shapes the denoising trajectory, but, via its influence on α_t , η_{t-1} , and η_t , also determines the relative weight of the training loss at each timestep:

$$\text{Loss coefficient: } \frac{\alpha_t}{2\kappa^2\eta_{t-1}\eta_t}$$

Intuitively, this schedule yields an *extremely large* loss coefficient at early (low-noise) timesteps—when the image is nearly clean—because errors here cannot be corrected later, so the model is heavily penalized for mistakes on almost-clean inputs. In contrast, the coefficient becomes very small at large (high-noise) timesteps, where the sample is highly corrupted and denoising is inherently more uncertain.

While this design makes sense from a theoretical perspective, the resulting extreme imbalance (very large coefficients at early steps) leads to training instability in practice. For this reason, following previous work, we omit the per-step coefficient and use a uniform loss weighting across timesteps during optimization.

3 Accelerating Inference via Consistency Distillation

While residual diffusion cuts the denoising process from roughly 1000 steps in standard DDPMs to as few as 15, those 15 steps are still expensive—about eight hours on an A100 GPU to generate 13,800 images. To accelerate inference further and make residual diffusion competitive with fast, non-diffusion approaches, we adopt *consistency models* [13].

Consistency distillation is a principled approach that allows diffusion models to generate high-quality outputs in dramatically fewer steps—potentially even in a single step—by aligning predictions across all timesteps. The central idea is to ensure that the model’s outputs at every denoising step are consistent, so that the final result does not depend on the particular reverse trajectory or Markov chain path. Importantly, residual diffusion satisfies all theoretical requirements for consistency distillation, making it a direct special case within the broader consistency model framework (see Section 3.1 for details). As a result, we can apply consistency distillation to our model with no architectural modifications.

Our motivation for adopting consistency distillation is twofold: beyond enabling much faster inference, we observe that predictions at different timesteps in residual diffusion often capture complementary features (see Figure 4). For example, intermediate steps may better preserve local structures, while later steps tend to remove artifacts but can also oversmooth details. This suggests that aligning intermediate predictions through consistency distillation may not only shorten the inference trajectory, but also combine the strengths of each step—potentially yielding outputs that are both higher quality and more robust. Thus, we hypothesize that consistency distillation offers gains in both efficiency and performance for residual diffusion.

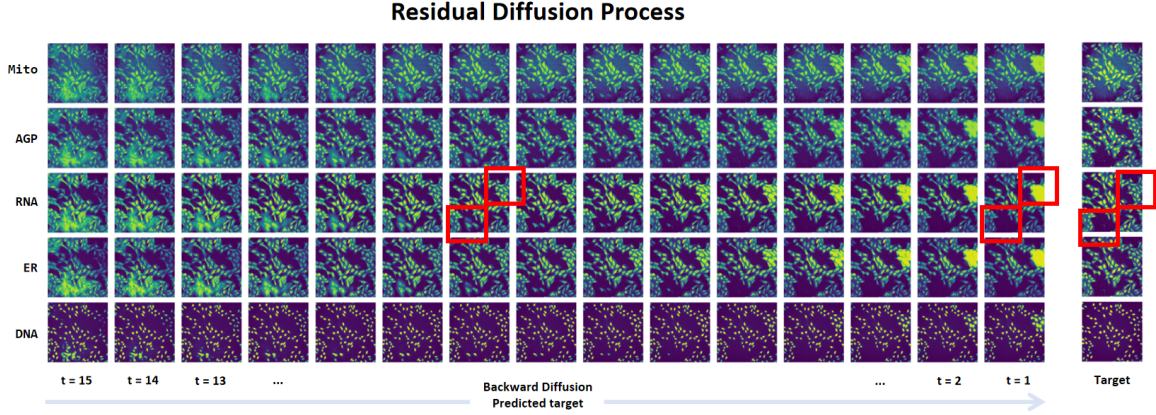


Figure 4: **Residual diffusion denoising over 15 reverse steps.** Columns show the model’s prediction of the clean image x_0 at each reverse-diffusion timestep ($t = 15$ on the left $\rightarrow t = 1$ on the right), with the ground-truth target in the final column. Rows correspond to the five Cell Painting channels (Mito, AGP, RNA, ER, DNA). In the RNA channel, red boxes highlight the difference between step 7 and step 1 compared to the target: step 7 reconstructs the upper-right region but introduces an artifact in the lower-left, while step 1 removes the artifact but slightly over-denoises the upper-right. This complementary error pattern motivates *consistency distillation*, which encourages intermediate and final predictions to agree and enables a shorter reverse trajectory.

3.1 Formal Definition of Consistency Models

A core property of consistency models is **self-consistency**: the model’s prediction should remain unchanged, regardless of the diffusion time-step along a given trajectory. In other words, at any point in the reverse process, the de-noised output for the clean image should be identical. This property allows the de-noising path to be dramatically shortened—even to a single step—since all intermediate predictions are aligned and thus redundant.

Figure 5 illustrates this concept. The model is trained so that its predictions at different time-steps (represented as arrows converging to a single point) always agree, no matter the specific path taken through the diffusion process.

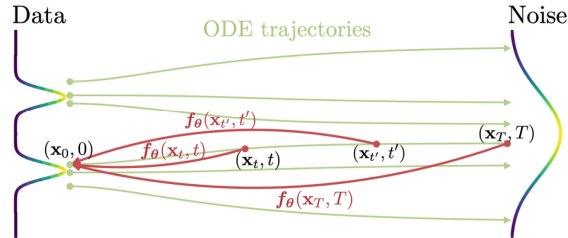


Figure 2: **Consistency models** are trained to map points on any trajectory of the PF ODE to the trajectory’s origin.

Figure 5: Illustration of the self-consistency property in consistency models. Regardless of the denoising trajectory or timestep, the model’s output converges to the same prediction, enabling step-agnostic inference.

Formally, following [13], let $\{x_t\}_{t \in [\epsilon, T]}$ denote the solution trajectory of the probability flow (PF)

ODE. A consistency function

$$f : (\mathbf{x}_t, t) \mapsto \mathbf{x}_\epsilon$$

is **self-consistent** if

$$f(\mathbf{x}_t, t) = f(\mathbf{x}_{t'}, t'), \quad \forall t, t' \in [\epsilon, T].$$

That is, the model predicts the same clean target image x_0 at every point along the trajectory, regardless of time-step. This behaviour is essential for efficient, step-agnostic inference.

To guarantee self-consistency, the model must also satisfy a *boundary condition*: it should act as the identity on nearly clean inputs,

$$f(\mathbf{x}_\epsilon, \epsilon) = \mathbf{x}_\epsilon,$$

so that already de-noised samples remain unchanged.

There are two common parametrizations that ensure this boundary condition:

- **Explicit Identity at Initial Time-step:**

$$f_\theta(\mathbf{x}, t) = \begin{cases} \mathbf{x}, & t = \epsilon \\ F_\theta(\mathbf{x}, t), & t > \epsilon \end{cases}$$

- **Skip-Connection Parametrization:**

$$f_\theta(\mathbf{x}, t) = c_{\text{skip}}(t) \mathbf{x} + c_{\text{out}}(t) F_\theta(\mathbf{x}, t),$$

where $c_{\text{skip}}(\epsilon) = 1$ and $c_{\text{out}}(\epsilon) = 0$.

In our residual diffusion model, the reverse process naturally follows the skip-connection parametrization and satisfies the boundary condition, aside from a negligible noise term. Specifically, at the initial step ($t = 1$), we have $\frac{\eta_{t-1}}{\eta_t} = 1$ and $\frac{\alpha_t}{\eta_t} = 0$, ensuring the model output exactly matches its input—thus realizing the identity mapping required for consistency models. Although our formulation includes a small noise term, our goal in cell image generation is deterministic translation between modalities. Therefore, when training consistency models, we omit the noise term in Equation 7, resulting in stable, reproducible outputs and full compatibility with the consistency framework.

Thus, the structure of our residual diffusion model—including its reverse process and noise scheduling—naturally meets both the theoretical and practical requirements for consistency modeling.

Remark on Noise Scheduling and Minimal Steps. It is important to note that, in principle, the noise schedule in consistency models may be chosen arbitrarily, and our particular schedule is fully compatible with the consistency framework. However, the specific choice of $\eta_1 \rightarrow 0$ (an almost noiseless starting point) and $\eta_T \rightarrow 1$ (a maximally noisy endpoint) introduces a practical constraint: it precludes true single-step denoising. The boundary condition at $\eta_1 \rightarrow 0$ yields a nearly clean starting sample, while $\eta_T \rightarrow 1$ corresponds to the most degraded state. As a result, at least two denoising steps are required to traverse the entire trajectory from source to target.

While this restriction does not strictly permit single-step prediction, it nonetheless enables a reduction in the number of denoising steps from 15 to only 2, representing a substantial acceleration over the original diffusion process.

On Adapting Multistep Sampling for Cell Images. For multi-step inference with consistency models, a greedy search to optimize the Fréchet Inception Distance (FID) metric across multi-step predictions is employed in the original paper. However, in the context of cell image generation, the preservation of morphological features and structural integrity is paramount—qualities that are not adequately captured by FID. Therefore, we adhere to the original residual diffusion reverse process rather than optimize for FID, as this approach is more consistent with the scientific objectives and qualitative priorities of cell imaging.

3.2 Consistency Distillation Loss

The central component of consistency distillation is its training objective, which enforces self-consistency across predictions at different timesteps. We adopt the loss function proposed in [13]:

Definition 1 (Consistency Distillation Loss). *The consistency distillation loss is defined as*

$$\mathcal{L}_{\text{CD}}(\theta, \theta^-; \varphi) := \mathbb{E} \left[\lambda(t_n) d(f_\theta(x_{t_{n+1}}, t_{n+1}), f_{\theta^-}(\hat{x}_\varphi^{t_n}, t_n)) \right],$$

where the expectation is taken over $x \sim p_{\text{data}}$, $n \sim U\{1, T-1\}$, and $x_{t_{n+1}} \sim \mathcal{N}(x; t_{n+1}I)$. Here, $U\{1, T-1\}$ denotes the uniform distribution over $\{1, 2, \dots, T-1\}$; $\lambda(\cdot) \in \mathbb{R}_+$ is a positive weighting function; and $d(\cdot, \cdot)$ is a non-negative metric, with $d(x, y) = 0$ if and only if $x = y$. The variable $\hat{x}_\varphi^{t_n}$ is defined as in Eq. (6) of [13]; θ^- denotes the exponential moving average (EMA) of past θ values, and f_θ is the current consistency model.

Algorithm 1 Consistency Distillation for Residual Diffusion

Require: Dataset $\mathcal{D}$, model parameters θ , learning rate η , noise schedule $\{\eta_t, \alpha_t\}$, distance metric $d(\cdot, \cdot)$, weighting function $\lambda(\cdot)$, EMA decay rate μ

- 1: Initialize EMA teacher: $\theta^- \leftarrow \theta$
 - 2: **while** not converged **do**
 - 3: Sample $x \sim \mathcal{D}$
 - 4: Sample $n \sim \mathcal{U}\{1, \dots, N-1\}$
 - 5: Sample $x_{t_{n+1}} \sim \mathcal{N}(x; \eta_{t_{n+1}}^2 I)$
 - 6: Compute $\hat{x}_\varphi^{t_n} \leftarrow \frac{\eta_{t_n}}{\eta_{t_{n+1}}} x_{t_{n+1}} + \frac{\alpha_{t_{n+1}}}{\eta_{t_{n+1}}} f_\varphi(x_{t_{n+1}}, t_{n+1})$
 - 7: Compute predictions:
 - 8: $y_{\text{student}} \leftarrow f_\theta(x_{t_{n+1}}, t_{n+1})$
 - 9: $y_{\text{teacher}} \leftarrow f_{\theta^-}(\hat{x}_\varphi^{t_n}, t_n)$
 - 10: Compute loss: $\mathcal{L} \leftarrow \lambda(t_n) d(y_{\text{student}}, y_{\text{teacher}})$
 - 11: Update model parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
 - 12: Update EMA teacher: $\theta^- \leftarrow \text{stopgrad}(\mu \theta^- + (1 - \mu) \theta)$
 - 13: **end while**
-

Intuitively, this loss encourages the student model f_θ to match the teacher’s predictions (parametrized by the EMA θ^-) at randomly chosen time-steps, thereby enforcing self-consistency throughout the de-noising trajectory. The metric $d(\cdot, \cdot)$ is flexible; in our experiments, we use a weighted combination of the *LPIPS* perceptual distance and mean squared error (MSE), consistent with the loss function employed in residual diffusion.

For the derivation of $\hat{x}_\varphi^{t_n}$, the original paper proposes

$$\hat{x}_\varphi^{t_n} := x_{t_{n+1}} + (t_n - t_{n+1}) \Phi(x_{t_{n+1}}, t_{n+1}; \varphi), \quad (15)$$

where Φ denotes the teacher model and serves as the update function for a one-step ODE solver applied to the empirical PF ODE. Since we employ a different noise schedule, we adapt this update using the rule in Equation 7 to recover the previous step. Here, we use a frozen pre-trained model to predict the update.

Therefore, training involves three models: the student f_θ , the teacher f_{θ^-} (an exponential moving average of the student), and an original reference model to generate $\hat{x}_\varphi^{t_n}$ for each batch. All models are initialized with the CellResDM pre-trained checkpoint. The student is updated at every step, the teacher is maintained as an EMA, and the fixed reference model guides the synthesis of intermediate samples. The entire training pipeline is outlined in Algorithm 1.

4 Evaluation Metrics

To rigorously evaluate the quality of image-to-image translation in our framework, we employ a comprehensive suite of metrics that assess both standard image fidelity and biological validity. While conventional image metrics—such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM)—quantify pixel-wise and perceptual similarity, they do not directly reflect preservation of biologically meaningful features. Therefore, we supplement these with domain-specific evaluation derived from single-cell morphological features, extracted via CellProfiler¹, to ensure the synthetic images are useful for downstream scientific applications.

4.1 Standard Image Quality Metrics

We begin with widely adopted measures of pixel-level and perceptual similarity:

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (X_i - \hat{X}_i)^2$$

where N is the number of pixels, X_i is the intensity of the i -th pixel in the ground truth, and $\hat{X}_i$ is the corresponding prediction. Lower MSE values indicate better average fidelity.

Peak Signal-to-Noise Ratio (PSNR):

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right)$$

where MAX_I is the maximum possible pixel value (e.g., 255 for 8-bit images). Higher PSNR reflects lower noise and better reconstruction.

Structural Similarity Index (SSIM):

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

with μ_x , μ_y the local means, σ_x^2 , σ_y^2 the variances, σ_{xy} the local covariance, and C_1 , C_2 constants for stability. SSIM, ranging from 0 to 1, captures perceptual structure including edges and textures, making it particularly sensitive to cell boundaries.

¹CellProfiler: <https://cellprofiler.org>

Limitations of FID and Inception Score for Cell Images Despite their popularity in generative modeling for natural images, the Fréchet Inception Distance (FID) and Inception Score (IS) are fundamentally ill-suited for cell microscopy applications. Both metrics rely on feature representations learned by Inception networks trained on natural image datasets (e.g., ImageNet), which do not reflect the structure, texture, or semantic content of fluorescence microscopy images. As a result, FID and IS may provide misleading assessments, failing to capture biologically relevant differences or structural fidelity critical in cell imaging. Moreover, FID aggregates information across entire datasets, obscuring per-image and per-cell differences vital for phenotypic analysis. Therefore, we do not report FID or IS in our experiments, focusing instead on metrics directly tied to scientific objectives.

4.2 Biological Feature via CellProfiler

To address the shortcomings of standard image metrics in biological contexts, we evaluate feature-level correspondence using CellProfiler, an open-source toolkit for cell segmentation and quantitative analysis. By extracting a high-dimensional vector of morphological, intensity, and texture features from both real and synthetic images, we can directly assess whether the synthesized immunofluorescence images preserve information essential for biological interpretation.

Feature Extraction. Our analysis uses a CellProfiler pipeline² based on [2], which segments nuclei and whole cells and computes a broad set of features per object. These include measurements of area, shape, granularity, intensity, texture, and spatial relationships, summarized in Table 1. Bright-field channels, not generated by our model, are excluded.

Table 1: Representative feature categories extracted by CellProfiler for evaluation.

Category	Example Features
Image Quality	Multiscale blur, saturation, intensity stats, thresholding
Colocalization	Pearson r , Manders', rank-weighted coefficients
Granularity	Spectra at multiple scales (up to 16 px)
Intensity	Mean, total, radial intensity, Zernike moments
Size & Shape	Area, perimeter, axis lengths, Zernike descriptors
Texture	GLCM-based texture (contrast, correlation, etc.)
Neighbors	Neighbor counts, contact fractions
Counts & QC	Cell/nucleus counts, QC flags

Overall, we extract 1,180 features for whole cells, 1,165 for cytoplasm, and 1,173 for nuclei per image. These descriptors form the basis for quantitative comparison between real and synthesized data.

4.2.1 Object Matching for Single-Cell Analysis

A critical challenge in quantitative biological evaluation is ensuring a meaningful correspondence between individual cells in real and synthetic images. Due to stochasticity in segmentation and noise, independently processed images may differ in both cell count and spatial arrangement, complicating direct comparison of per-cell features.

²Pipeline available at https://github.com/jump-cellpainting/2024_Chandrasekaran_NatureMethods/tree/main/pipelines/2020_11_04_CPJUMP1

Motivation. Figure 6 demonstrates that, even when real and generated images are visually similar, automated segmentations can yield different cell boundaries and counts. Thus, an explicit matching step is required to align cells (or nuclei) before computing feature correlations.

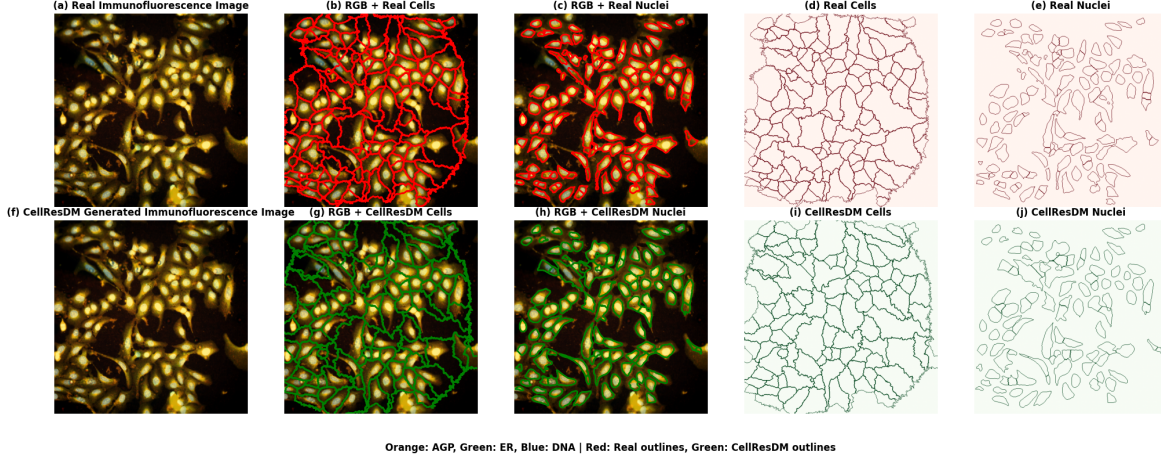


Figure 6: **CellProfiler segmentations on real and CellResDM-generated immunofluorescence images.** (a, f): Raw images; (b, g): boundaries; (c, h): nuclei; (d, i), (e, j): segmentation outlines for cells/nuclei, real (top) and synthetic (bottom). Red: real segmentations; Green: CellResDM segmentations. Despite high visual similarity, segmentation outcomes differ, motivating the need for explicit object matching.

Matching Algorithms. To establish per-object correspondences, we use two matching strategies:

- **Centroid Matching:** Each synthetic cell is paired with the nearest real cell by Euclidean centroid distance on their center, subject to a maximum distance threshold.
- **IoU Matching:** The intersection-over-union (IoU) between all real-synthetic mask pairs is computed, and the optimal one-to-one assignment is found via the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`), retaining only matches above an IoU threshold.

IoU matching, while more computationally demanding, yields more reliable pairs in dense or morphologically variable settings and serves as our primary method. Centroid matching provides a fast alternative for large-scale evaluation.

4.2.2 Feature Correlation Computation

With one-to-one cell correspondences established, we quantify how well morphological features are preserved by computing the Pearson correlation coefficient for each feature across all matched pairs between real and each synthetic method’s produced image. For feature j , let $\{f_{ij}^{(\text{real})}\}$ and $\{f_{ij}^{(\text{syn})}\}$ denote the values of feature j for matched cell i in the real and synthetic images, respectively. We then compute

$$r_j = \text{corr} \left(\{f_{ij}^{(\text{real})}\}, \{f_{ij}^{(\text{syn})}\} \right).$$

Repeating this for all features and compartments (cell, nucleus) yields a distribution of correlation values for each method. We summarize performance by reporting the mean and standard deviation of these correlations, providing a quantitative measure of single-cell morphological fidelity.

4.2.3 Unsupervised Feature Space Visualization

Beyond per-feature correlations, we qualitatively assess the preservation of global morphological structure using unsupervised dimensionality reduction. Specifically, we jointly embed the high-dimensional CellProfiler feature vectors from both real and synthesized images into a two-dimensional space via Uniform Manifold Approximation and Projection (UMAP) [11]. By visualizing the combined feature distributions, we can directly evaluate whether synthetic cells occupy similar regions of morphological feature space as real cells, and whether the diversity of cellular phenotypes is faithfully reproduced.

Meaningful overlap between the real and synthetic point clouds in UMAP space indicates that the generative model successfully captures the complex, population-level variability and inter-feature relationships observed in real data. Conversely, systematic shifts, fragmentation, or mode collapse in the synthetic distribution may highlight areas where biological fidelity is lacking. This qualitative visualization thus complements the quantitative correlation analysis, providing an intuitive assessment of overall model performance in recapitulating cellular heterogeneity.

4.3 Summary of Evaluation Protocol

By combining standard image quality metrics with detailed, cell-level biological feature analysis, our evaluation protocol ensures that models are judged not only by their ability to reconstruct images at the pixel and perceptual level, but also by their capacity to preserve information critical for scientific discovery. This dual approach distinguishes our work from most prior art in generative modeling, where evaluation often stops at global metrics ill-suited for biological applications.

In practice, we report all metrics across held-out test images and provide ablation results by compartment (nucleus, cytoplasm, whole cell) and feature category, offering transparency into which biological properties are best retained and where errors remain. This multidimensional evaluation is essential for building trust in label-free predictive imaging as a tool for high-content screening and phenotypic profiling.

5 Dataset and Experimental Setting

5.1 Dataset

Our experiments utilize the public **cpg0000** dataset from the Cell Painting Gallery [2], comprising over three million paired brightfield (BF) and immunofluorescence (IF) images. For robust evaluation, we curated a representative subset of eight plates, stratified by perturbation type and cell line (Table 2). Of these plates, four were used for training, two for validation, and two for testing. To assess the generalisability of our models, two additional plates under distinct CRISPR perturbations (marked with *) were reserved exclusively for out-of-distribution evaluation.

Imaging Modalities.

- **Brightfield (BF):** Each field of view is captured at three z -planes: the central focal plane (Brightfield), $5\text{ }\mu\text{m}$ above (BFHigh), and $5\text{ }\mu\text{m}$ below (BFLow). WE use only the central Brightfield channel in our main experiments to ensure simplicity and comparability with prior work.
- **Immunofluorescence (IF)/Cell Painting:** Five fluorescent channels highlight distinct cellular components according to the Cell Painting protocol: AGP (actin, Golgi, plasma membrane), DNA (nucleus), ER (endoplasmic reticulum), Mito (mitochondria), and RNA (nucleoli and cytoplasmic RNA).

Plate Barcode	Perturbation	Cell Type	Num. Images	Purpose
BR00116991	compound	A549	27,648	Train
BR00116992	compound	A549	27,640	Train
BR00116995	compound	U2OS	27,648	Train
BR00117024	compound	U2OS	27,648	Train
BR00116993	compound	A549	27,352	Validation
BR00117025	compound	U2OS	27,648	Validation
BR00116994	compound	A549	27,576	Test
BR00117026	compound	U2OS	27,648	Test
BR00118041*	crispr	A549	27,560	OOD Test
BR00118045*	crispr	U2OS	27,648	OOD Test

Table 2: Summary of plates and data splits from the cpq0000 dataset. Plates marked with an * were reserved exclusively for out-of-distribution (OOD) generalization evaluation.

5.2 Experimental Setting

Model Architecture. Our core architecture is a U-Net augmented with Swin Transformer modules (UNetModelSwin). The model processes concatenated brightfield and noised-residual inputs (each resized to 512×512 pixels) and predicts the five-channel immunofluorescence output. Key architectural choices include 160 base channels, spatial attention at resolutions $\{64, 32, 16, 8\}$, two residual blocks per stage, Swin layers with depth 2 and 192-dimensional embeddings, window size 8, and MLP ratio 4.

Noise Scheduling and Training Objective. We train our diffusion model with the non-uniform geometric noise schedule described in Section 2.3 ($\kappa = 2.0$, $p = 0.3$, $\eta_T = 0.99$, $T = 15$ steps). To stabilize optimization, we omit the explicit loss coefficient and supervise learning with a weighted combination of mean squared error (MSE) and perceptual LPIPS [18] losses, as detailed in Section 2.2.

Data Preprocessing. Prior to training and inference, all images are uniformly resized to 512×512 pixels. Contrast is enhanced by clipping the lowest and highest 5% of pixel intensities in each image. Pixel values are normalized to $[0, 1]$ for training, while $[0, 255]$ scaling is used for visualization and final evaluation.

Baselines. We benchmark our approach against three complementary baselines:

- **U-Net** [12]: A widely adopted, deterministic baseline for image-to-image translation.
- **Pix2Pix** [7]: A conditional GAN, outperforming more advanced architectures (e.g., SPADE, SPADE-Diffusion, PLATTE) for Cell Painting synthesis as shown in [16].
- **DeepHCS** [10]: A two-stage network designed for immunofluorescence synthesis, where the first stage translates brightfield to immunofluorescence, and the second stage denoises the prediction.

These baselines together span deterministic, generative, and task-specific generative modeling approaches.

Ablation Studies. To dissect the contribution of each modelling component, we perform three ablation test:

1. *LPIPS Loss:* We retrain with only MSE supervision to assess the benefit of perceptual loss.
2. *Loss Coefficient:* We reintroduce the timestep-dependent loss coefficient to measure its impact on training stability and image quality.
3. *Latent Diffusion:* We encode five-channel immunofluorescence images with a pretrained SD-VAE, splitting them into two three-channel groups, and train the diffusion process in this latent space to evaluate effects on reconstruction fidelity.

Consistency Distillation. To accelerate inference, we apply consistency distillation to our fully trained residual diffusion model. Fine-tuning is performed for one epoch with an EMA decay of 0.9999. At inference, we made the distilled model use only two reverse-diffusion steps.

Training Protocol. All experiments use a fixed seed of 42, learning rate 5×10^{-5} , cosine learning-rate schedule with 5,000 warmup steps, and are run for 10 epochs. Batch size is 2 per device for training and 4 for evaluation; no weight decay is used.

6 Experimental Results

In this section, we present a comprehensive evaluation of our proposed methods and baselines, focusing on both quantitative metrics and cell feature preservation.

6.1 quantitative Results

We first assess the pixel-wise and perceptual accuracy of each model using the standard metrics described in Section 4: MSE, PSNR, SSIM. Each metric is computed independently for all five immunofluorescence channels, with results aggregated across the entire test set.

The detailed results are summarized in Table 3 and visualized as channel-wise distributions in Figure 7.

Baseline Performance. Among all baselines, **DeepHSC** achieves the highest overall quantitative scores, outperforming both the U-Net and Pix2Pix models as well as the original CellResDM in terms of lower MSE and higher SSIM/PSNR. While it is somewhat unexpected for DeepHSC to marginally outperform CellResDM, the performance gap remains modest.

Effect of Consistency Distillation. Notably, the **Consistency Distilled CellResDM** (2-step) variant achieves a substantial performance boost over all other models, recording the lowest average MSE and the highest SSIM and PSNR across most channels. This demonstrates that consistency distillation not only accelerates inference but also enhances reconstruction fidelity, consistent with our hypothesis.

Channel-wise Trends. Across all models, the DNA channel consistently yields the best reconstruction metrics—highest SSIM and PSNR, and lowest MSE. This advantage arises because nuclear staining results in compact, spatially localized “blobs” that are distinctly separated for each cell, making nuclei morphologically simple and straightforward to predict. In contrast, the Mito, AGP, RNA,

and ER channels capture a range of cytoplasmic and organelle features, as well as some nuclear details. These compartments are inherently more complex: they exhibit highly textured patterns, diffuse and sometimes overlapping boundaries, and significant heterogeneity both within and across cells. As a result, these channels present a much greater challenge for accurate synthesis and morphological fidelity in all models.

Model	MSE ₂₅₅ ↓	SSIM ₂₅₅ ↑	PSNR ₂₅₅ ↑
DeepHSC	617.9 ± 473.6	0.794 ± 0.103	20.85 ± 2.14
U-Net	1140.5 ± 1908.3	0.622 ± 0.101	18.99 ± 2.69
Pix2Pix	1626.1 ± 829.5	0.433 ± 0.087	16.44 ± 1.83
CellResDM (15 steps)	734.6 ± 543.3	0.728 ± 0.119	20.24 ± 2.49
CellResDM (2 steps, Consistency Distilled)	568.7 ± 1136.1	0.768 ± 0.126	21.98 ± 2.58

Table 3: Mean ± standard deviation of quantitative metrics across the test set for each model. Lower MSE and higher SSIM/PSNR indicate improved reconstruction quality.

6.2 Ablation Results

To clarify the necessity of each component in our framework, we did ablation study on three aspects of CellResDM: (i) the LPIPS loss, (ii) omission of the loss coefficient, and (iii) latent diffusion via SD-VAE. Figure 8 and Table 4 summarize the results. In every case, these ablations significantly degrade performance relative to the baseline, confirming the importance of each design choice.

Model	MSE ₂₅₅ ↓	SSIM ₂₅₅ ↑	PSNR ₂₅₅ ↑
CellResDM (15 step)	734.6 ± 543.3	0.728 ± 0.119	20.24 ± 2.49
CellResDM (Consistency Distilled)	568.7 ± 1136.1	0.768 ± 0.126	21.98 ± 2.58
CellResDM (no LPIPS)	1573.8 ± 803.0	0.549 ± 0.100	16.65 ± 2.06
CellResDM (with loss coefficient)	878.0 ± 509.0	0.689 ± 0.108	19.16 ± 1.91
CellResDM (SD-VAE latent diffusion)	1451.3 ± 521.5	0.475 ± 0.089	16.74 ± 1.39

Table 4: Mean ± standard deviation of metrics for each ablation variant (lower MSE, higher SSIM/PSNR are better).

LPIPS Loss. Omitting LPIPS from the loss dramatically worsens all metrics. MSE alone penalizes only pixel-wise errors, often producing blurry outputs and failing to capture high-frequency perceptual details. Although LPIPS is trained on natural images, it encourages similarity in deep feature space, guiding the network to preserve cellular texture and structure—resulting in reconstructions that are sharper and more realistic to the human eye.

Loss Coefficient. When reintroduced, the imbalanced loss coefficient produces unstable training, as evidenced by the loss curve in Figure 9, and consistently degrades reconstruction quality.

Latent Diffusion. We further evaluated CellResDM in a latent diffusion setting using a pretrained SD-VAE encoder³. SD-VAE can reconstruct cell images with very low MSE and no fine-tuning (Figure 10), confirming its suitability for the domain. However, applying residual diffusion in latent

³SD-VAE model weight: <https://huggingface.co/stabilityai/sd-vae-ft-mse>

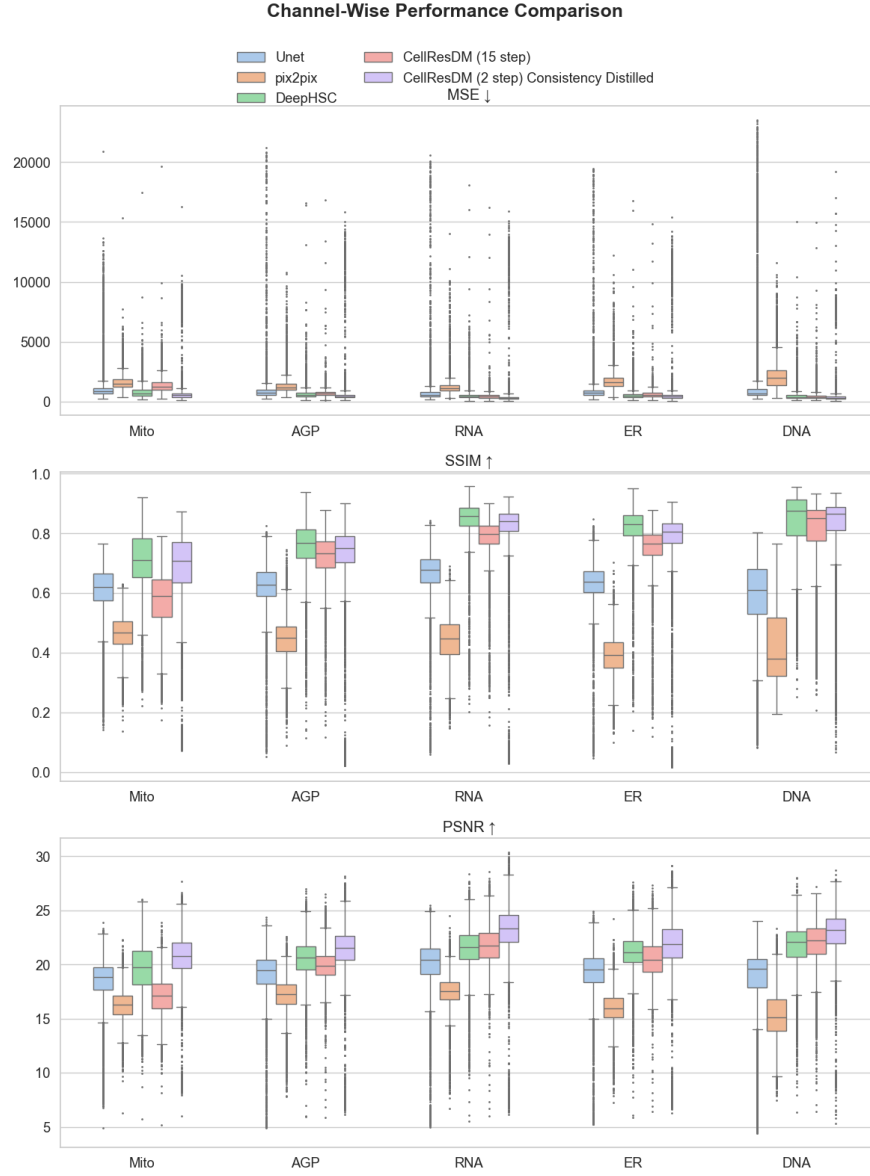


Figure 7: **Channel-wise comparison of reconstruction metrics** (MSE, SSIM, PSNR) for all evaluated models. Boxplots display the distribution for each channel (Mito, AGP, RNA, ER, DNA) over the test set. DeepHSC, CellResDM, and Consistency Distilled CellResDM consistently outperform other baselines, with the DNA channel being the most reliably reconstructed.

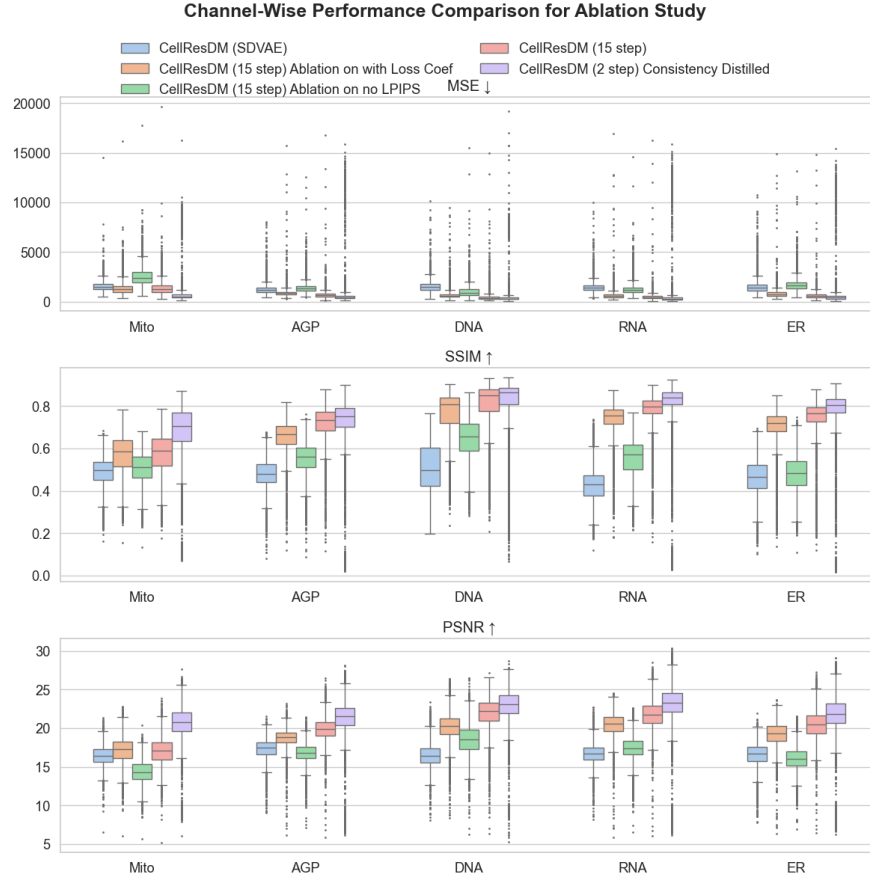


Figure 8: Quantitative ablation results for five imaging channels (Mito, AGP, DNA, RNA, ER). Boxplots show MSE (lower is better), SSIM, and PSNR (both higher are better). All ablated variants underperform both baseline and consistency-distilled CellResDM.

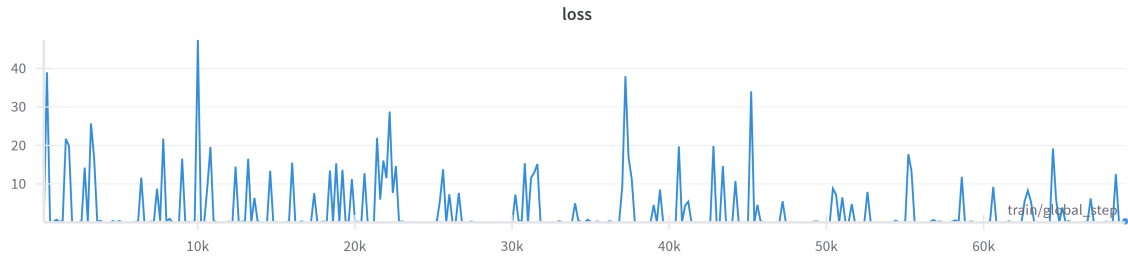


Figure 9: Loss curve for CellResDM with reintroduced loss coefficient, showing instability and poor convergence.

space resulted in much poorer performance. This aligns with prior observations [16], likely because the linear residual shift in pixel space becomes a complex, nonlinear mapping in the SD-VAE latent space, substantially complicating optimization and reducing fidelity.

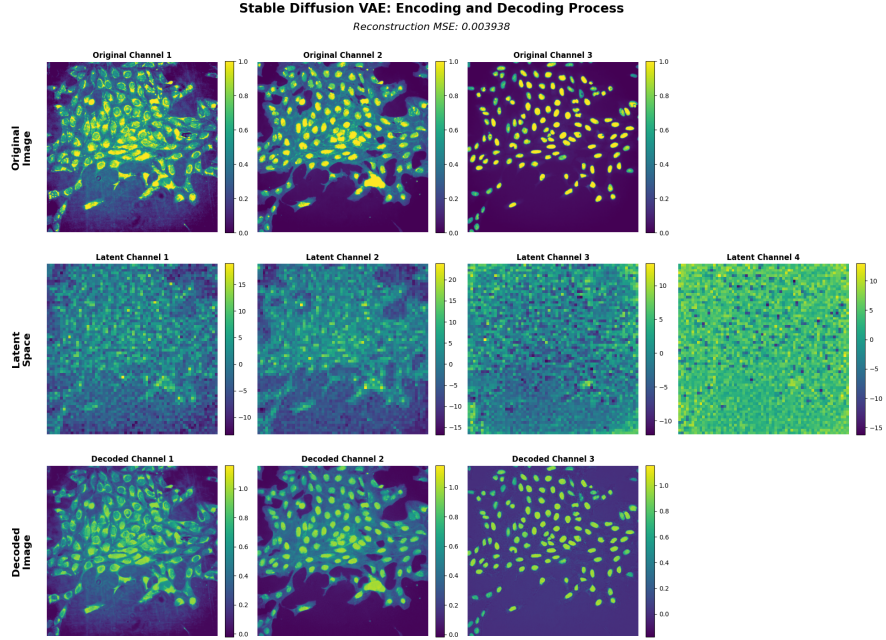


Figure 10: SD-VAE reconstructions (bottom row) of cell images (top row) closely match the originals, but latent diffusion fails to achieve similar fidelity.

6.3 Qualitative Results

We present a qualitative comparison of real and synthetic images for all models in Figures 11 and 12.

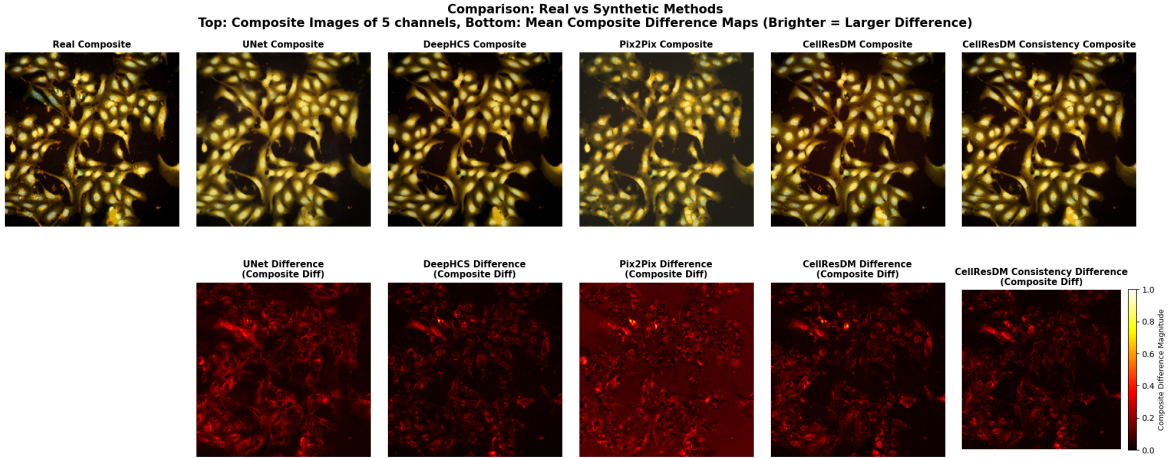


Figure 11: Top: Composite images of all five channels for real and synthetic data. Bottom: Mean difference maps (brighter = greater deviation from real). CellResDM and its consistency-distilled variant show the smallest errors, indicating high fidelity.

As shown in Figure 11, Unet outputs appear blurry with weak cell boundaries, while Pix2Pix is high in error. DeepHCS, CellResDM and Consistency Distilled CellResDM stand out for their close visual match to real images, as evidenced by minimal difference in the error maps.

Figure 12 provides zoomed-in views to highlight fine structural details. Unet’s outputs are over-

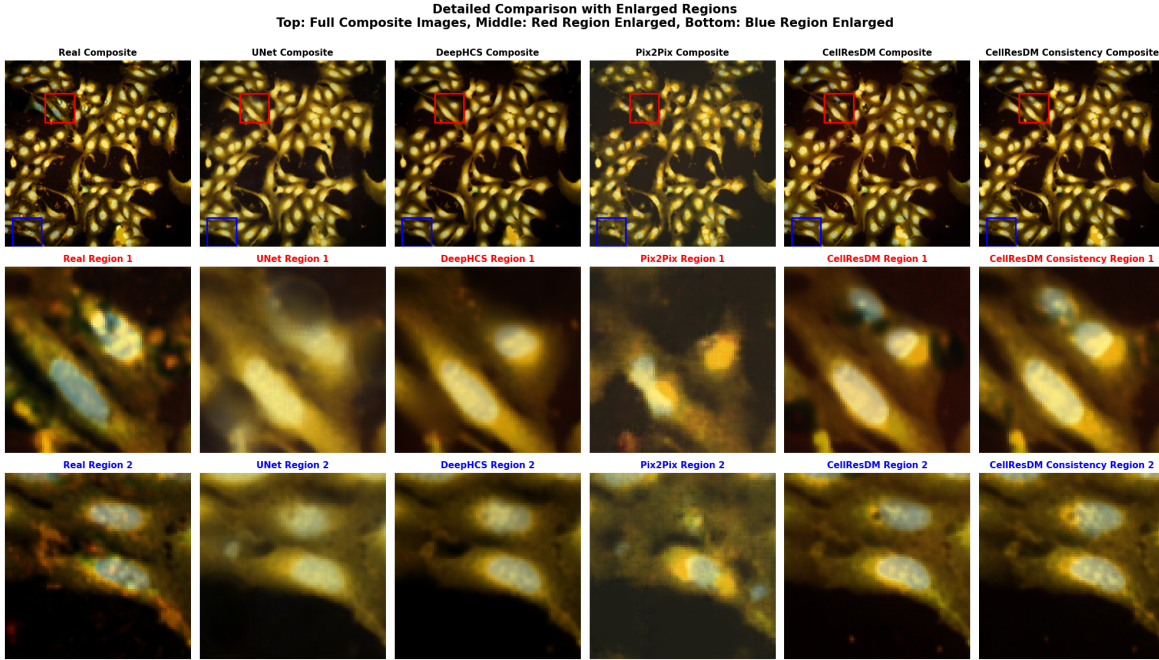


Figure 12: Enlarged qualitative comparison. Top: Full composites; middle/bottom: Zoom-ins of red- and blue-boxed regions. CellResDM and its consistency-distilled variant best preserve fine cellular structures; baselines often lose detail or introduce artifacts.

smoothed, losing key nuclear features and cell boundaries. Pix2Pix frequently hallucinates or erases cellular structures, leading to inconsistent results. DeepHCS often omits portions of cells and still produces overly smoothed morphologies. In contrast, both CellResDM and its consistency-distilled variant accurately reconstruct both global morphology and fine cellular detail, consistently matching the real images even in the most challenging regions.

These qualitative results reinforce the quantitative findings: CellResDM and its consistency-distilled variant offer the highest visual fidelity and structural accuracy among all evaluated methods.

6.4 Morphological Feature Fidelity

While high-level image similarity metrics provide a useful baseline, they do not guarantee that synthetic images retain biologically meaningful structure at the single-cell level. To directly assess the biological realism of each synthesis model, we quantified the preservation of *single-cell morphological features* extracted using CellProfiler. Correlations were computed between matched real-synthetic object pairs (Sec. 4.2.1), summarizing how faithfully each method preserves cell-level phenotypes.

Global performance. Figure ?? visualizes feature correlations by compartment; Table 5 summarizes the results for each model. The consistency-distilled **CellResDM** achieves the highest overall correlation ($r = 0.565$), narrowly surpassing **DeepHCS** ($r = 0.555$) and the original 15-step **CellResDM** baseline. **UNet** trails with lower correlations, and **Pix2Pix** performs worst ($r = 0.314$), consistent with its qualitative artifacts (Fig. 12).

Feature-specific trends.

Table 5: Mean Pearson r by compartment (numeric summary of Fig. 13). Higher is better.

Model	Cell	Cytoplasm	Nuclei	Overall
UNet	0.434	0.401	0.413	0.416
DeepHCS	0.572	0.542	0.552	0.555
Pix2Pix	0.327	0.280	0.335	0.314
CellResDM	0.527	0.494	0.574	0.532
CellResDM_Distill	0.558	0.530	0.607	0.565

- **Relational (Parent) features:** All models achieve near-perfect fidelity ($r > 0.95$) in assigning objects (e.g., nuclei within cells). This indicates that, despite intensity or texture errors, all nuclei remain correctly nested within their corresponding cell. Thus, no systematic or readily observable biological errors occur at the object-assignment level across any method.
- **Count-based (Number, Neighbors) features:** Correlations range from $r \approx 0.4$ – 0.7 , scaling with overall model quality. Pix2Pix underestimates cell and nucleus counts due to missing/merged objects, while UNet’s over-smoothing affects neighbor relationships. Diffusion-based models, especially consistency-distilled CellResDM, achieve the highest accuracy, though tightly clustered doublets remain challenging.
- **Spatial statistics (Location, Correlation):** Features capturing geometry or long-range spatial dependencies remain difficult, with no model exceeding $r = 0.40$. Pix2Pix again performs worst ($r < 0.25$), and even the best models struggle to reproduce radial or texture patterns, reflecting limitations of current loss functions for enforcing higher-order structure.

6.5 Channel-wise Texture Feature Correlations

To better understand the preservation of morphological detail, we evaluated model performance across four families of CellProfiler-derived texture features—*Granularity*, *Intensity*, *Radial Distribution*, and *Texture*—for each of the five Cell Painting channels (Mito, AGP, RNA, ER, DNA) and all three compartments (cell, cytoplasm, nucleus). Figure 14 presents the Pearson correlation coefficients (r) for each combination.

Intensity features: First-order intensity statistics are the most robustly captured, with median correlations $r \approx 0.7$ for CellResDM, DeepHCS, and their distilled variants, and $r \approx 0.5$ – 0.6 for UNet. Pix2Pix consistently lags behind, with DNA channel performance particularly poor. These results confirm that pixel-based and perceptual losses are well-suited to preserving overall brightness and contrast.

Granularity features: Granularity, which quantifies the presence of small, repetitive structures, shows improved correlation with more advanced architectures. While Pix2Pix achieves $r \approx 0.2$, this rises to 0.3 in UNet, and 0.4 or above for DeepHCS and diffusion-based models—especially in mitochondrial and RNA channels. However, cytoplasmic granularity remains more challenging, with generally lower scores, reflecting difficulties in capturing subtle, low-contrast textures.

Radial distribution: All models struggle with radial distribution features, which encode concentric intensity gradients and spatial organization. Even the best models rarely exceed $r \approx 0.5$ in the cell

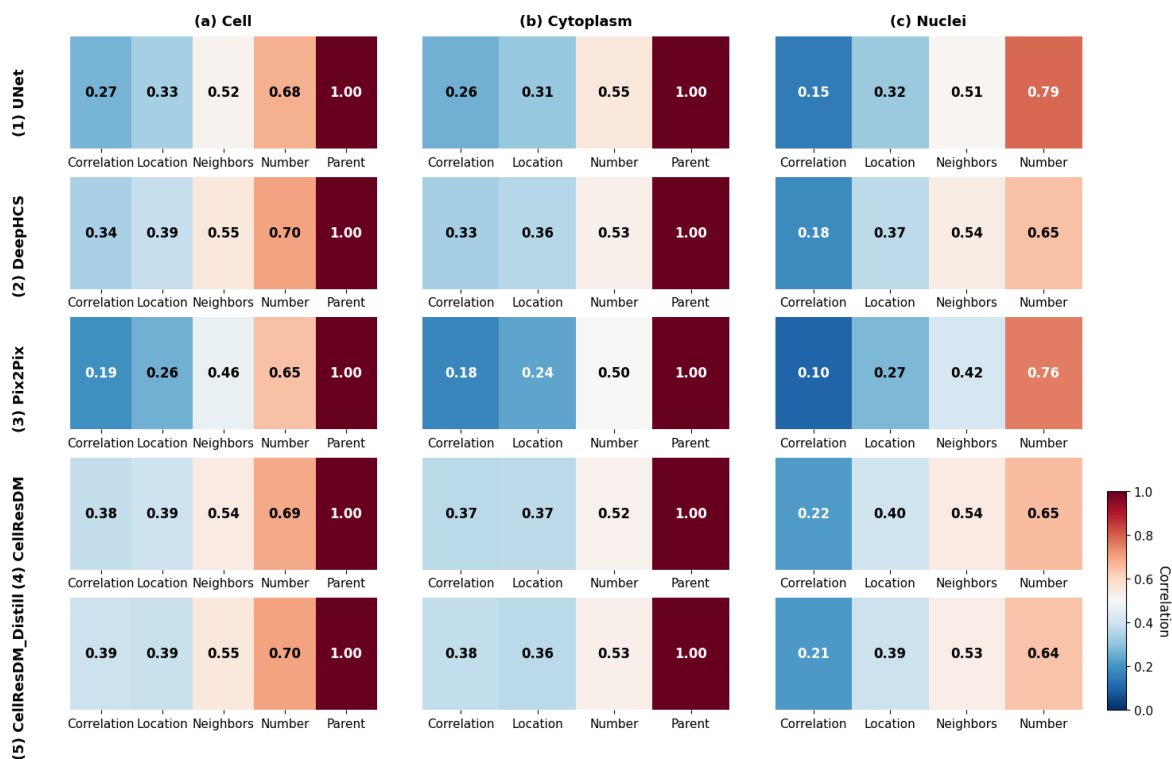


Figure 13: Correlation matrix for morphological feature groups across cell, cytoplasm, and nuclei compartments.

and nucleus, and fall below 0.4 in cytoplasm. Pix2Pix again performs the worst, seldom surpassing $r = 0.3$. These results highlight the limitations of current architectures and loss functions in modeling long-range spatial dependencies.

Texture features fall between the above extremes. CellResDM, DeepHCS, and their distilled versions achieve correlations in the 0.5–0.7 range, with UNet somewhat lower and Pix2Pix lowest overall. Notably, the DNA channel, while easy for intensity, offers less advantage for higher-order texture metrics, likely due to the simpler spatial structure of nuclear staining.

Across all texture feature families and imaging channels, the same ranking of model performance consistently emerges: Pix2Pix performs worst, followed by UNet, then CellResDM, with DeepHCS achieving slightly better results, and the consistency-distilled CellResDM variant achieving the highest correlations overall. The distilled two-step CellResDM generally matches or outperforms CellResDM, confirming that efficiency gains do not come at the cost of biological fidelity.

Taken together, these findings demonstrate that while modern diffusion and transformer-based methods reproduce first-order and medium-scale texture statistics well, further advances are needed to fully capture higher-order spatial and organizational features in cellular images.

6.6 UMAP Visualization of Morphological Feature Distributions

While feature correlations quantify cell-level fidelity, they do not capture the global structure of the feature distributions. To visualize this, we jointly embed CellProfiler feature vectors from real and

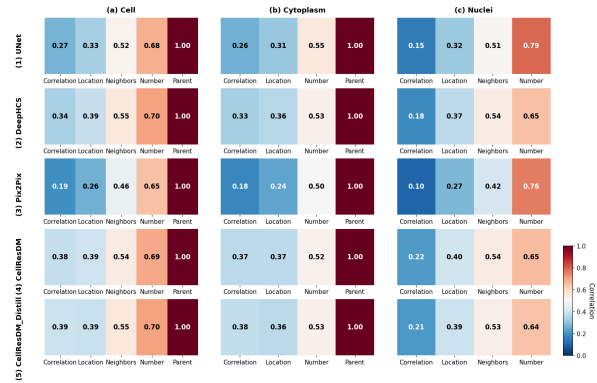


Figure 14: Correlation matrix for morphological feature groups across cell, cytoplasm, and nuclei compartments.

synthetic images using UMAP [11].

Figure 15 shows UMAP projections for whole-cell, cytoplasmic, and nuclear features. Each point represents a cell or nucleus colored by synthesis method. Strong overlap with the real distribution indicates high morphological realism.

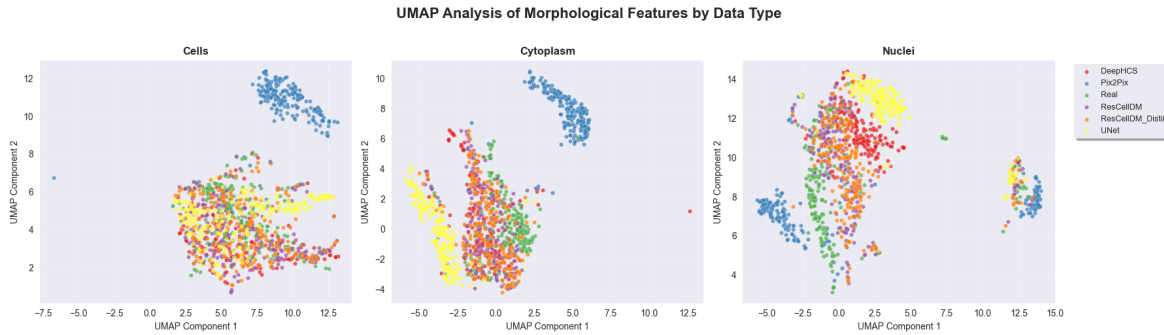


Figure 15: UMAP projections of CellProfiler features for whole cells, cytoplasm, and nuclei. Each point is a cell or nucleus from real or synthetic images, colored by method. Overlap indicates preservation of population-level morphology.

In all three compartments, **CellResDM** and its consistency-distilled variant show near-complete overlap with the real data, faithfully capturing the full diversity of cellular morphologies. **DeepHCS** also aligns closely but cluster slightly above the main real-cell manifold in the nucleus panel, indicating minor deviations in nuclear detail. **UNet** exhibits noticeable shifts—particularly toward higher UMAP₁—and broader dispersion, reflecting its tendency to over-smooth boundaries and blur textures. **Pix2Pix** forms distinct, peripheral clusters separated from the real distribution, consistent with its frequent loss or misplacement of cells and structures. Overall, these patterns reinforce that only CellResDM and Consistency Distilled CellResDM generate synthetic populations that truly mirror the morphological complexity of real cells, while others distort population structure and reduce biological fidelity.

7 Discussion and Conclusion

This work reproduced CellResDM, a residual-shifted diffusion framework for brightfield to immunofluorescence image synthesis, and demonstrates its superior performance compared to established baselines. Our comprehensive evaluation shows that CellResDM, especially when combined with consistency distillation, outperforms DeepHCS, Pix2Pix, and UNet both quantitatively and in biologically meaningful terms. The model not only achieves lower MSE and higher SSIM/PSNR but also excels at preserving morphological features, as confirmed by detailed correlation analysis and UMAP visualization.

Consistency distillation proves especially impactful: reducing inference from 15 to just 2 steps yields significant acceleration without sacrificing fidelity—indeed, we often observe slight improvements in both global image quality and morphological preservation. Our ablation studies further reveal that each component of the method is necessary; perceptual loss, balanced noise scheduling, and direct pixel-space modelling each contribute to stable optimization and faithful reconstruction.

Nonetheless, certain limitations persist. Capturing fine-grained texture and spatial distribution features, such as radial gradients or higher-order texture cell features, remains challenging, indicating room for improvement in future architectures or loss designs. Furthermore, although CellResDM generalizes well to held-out and out-of-distribution conditions within the current dataset, evaluation on rarer cell types or diverse imaging protocols will be essential to fully validate robustness.

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A Mathematical Derivations for CellResDM

In this appendix, we provide detailed mathematical derivations and proofs that underpin the main methods described in Section 2. These include the reparameterization of the forward process by induction, the computation of the true posterior for the reverse process, and the derivation of the per-step KL loss coefficient for training. Readers less interested in technical derivations may safely skip to the main results, but these details may be useful for those seeking to reproduce or extend our work.

A.1 Forward Process Reparameterization by Induction

As described in Section 2, the forward noising process can be reparameterized to depend directly on the initial images x_0 and y_0 . We now provide the induction proof supporting Equation (2).

Base Case ($t = 1$):

$$\begin{aligned} q(x_1|x_0, y_0) &= \mathcal{N}(x_1; x_0 + \alpha_1 e_0, \kappa^2 \alpha_1 I) \\ &= \mathcal{N}(x_1; x_0 + \eta_1 e_0, \kappa^2 \eta_1 I) \quad (\text{since } \alpha_1 = \eta_1) \end{aligned}$$

which matches Eq. (2) for $t = 1$.

Inductive Step: Assume $q(x_t|x_0, y_0) = \mathcal{N}(x_t; x_0 + \eta_t e_0, \kappa^2 \eta_t I)$ holds for some t . Then,

$$\begin{aligned} x_{t+1} &= x_t + \alpha_{t+1} e_0 + \kappa \sqrt{\alpha_{t+1}} \epsilon_{t+1} \\ &= (x_0 + \eta_t e_0 + \kappa \sqrt{\eta_t} \epsilon_t) + \alpha_{t+1} e_0 + \kappa \sqrt{\alpha_{t+1}} \epsilon_{t+1} \\ &= x_0 + (\eta_t + \alpha_{t+1}) e_0 + \kappa \sqrt{\eta_t} \epsilon_t + \kappa \sqrt{\alpha_{t+1}} \epsilon_{t+1} \\ &= x_0 + \eta_{t+1} e_0 + \kappa \sqrt{\eta_t} \epsilon_t + \kappa \sqrt{\alpha_{t+1}} \epsilon_{t+1} \end{aligned}$$

The sum of two independent Gaussians is Gaussian with variance $\kappa^2(\eta_t + \alpha_{t+1})I = \kappa^2 \eta_{t+1}I$, thus:

$$x_{t+1} \sim \mathcal{N}(x_0 + \eta_{t+1} e_0, \kappa^2 \eta_{t+1} I)$$

completing the induction.

A.2 True Posterior for the Reverse Process

The main text (Section 2) gives the form of the true reverse posterior, which is essential for deriving the training objective. Here, we show its derivation by completing the square for two Gaussian distributions.

Recall:

$$\begin{aligned} q(x_t|x_{t-1}, y_0) &= \mathcal{N}(x_t; x_{t-1} + \alpha_t e_0, \kappa^2 \alpha_t I) \\ q(x_{t-1}|x_0, y_0) &= \mathcal{N}(x_{t-1}; x_0 + \eta_{t-1} e_0, \kappa^2 \eta_{t-1} I) \end{aligned}$$

By Bayes' theorem and the Markov property, the true reverse posterior is:

$$\begin{aligned} q(x_{t-1}|x_t, x_0, y_0) &= \frac{q(x_t|x_{t-1}, y_0) q(x_{t-1}|x_0, y_0)}{q(x_t|x_0, y_0)} \\ &\propto q(x_t|x_{t-1}, y_0) q(x_{t-1}|x_0, y_0) \end{aligned}$$

Multiplying these two Gaussians (with respect to x_{t-1}) and completing the square, we obtain:

$$q(x_{t-1}|x_t, x_0, y_0) = \mathcal{N}(x_{t-1}; \mu_t, \Sigma_t)$$

with

$$\begin{aligned} \mu_t &= \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} x_0 \\ \Sigma_t &= \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I \end{aligned}$$

where $\eta_t = \eta_{t-1} + \alpha_t$.

A.3 KL Loss Coefficient Derivation

The per-step training loss for CellResDM is based on the KL divergence between the true posterior and the model's reverse step (see Section 2). Here, we expand the derivation.

Let:

$$\begin{aligned} q(x_{t-1}|x_t, x_0, y_0) &= \mathcal{N}(x_{t-1}; \mu_q, \Sigma_t) \\ p_\theta(x_{t-1}|x_t, y_0) &= \mathcal{N}(x_{t-1}; \mu_\theta, \Sigma_t) \end{aligned}$$

where

$$\begin{aligned} \mu_q &= \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} x_0 \\ \mu_\theta &= \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} f_\theta(x_t, y_0, t) \end{aligned}$$

The KL divergence between two Gaussians with the same covariance is:

$$D_{\text{KL}}(q(\mu_q, \Sigma_t) \| p_\theta(\mu_\theta, \Sigma_t)) = \frac{1}{2} (\mu_q - \mu_\theta)^\top \Sigma_t^{-1} (\mu_q - \mu_\theta)$$

Computing the mean difference:

$$\mu_q - \mu_\theta = \frac{\alpha_t}{\eta_t} (x_0 - f_\theta(x_t, y_0, t))$$

and

$$\Sigma_t = \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I \implies \Sigma_t^{-1} = \frac{1}{\kappa^2} \frac{\eta_t}{\eta_{t-1}} \frac{1}{\alpha_t} I$$

Plugging these in:

$$\begin{aligned} D_{\text{KL}} &= \frac{1}{2} \left(\frac{\alpha_t}{\eta_t} (x_0 - f_\theta) \right)^\top \frac{1}{\kappa^2} \frac{\eta_t}{\eta_{t-1}} \frac{1}{\alpha_t} \left(\frac{\alpha_t}{\eta_t} (x_0 - f_\theta) \right) \\ &= \frac{1}{2\kappa^2} \frac{\alpha_t}{\eta_{t-1}\eta_t} \|x_0 - f_\theta(x_t, y_0, t)\|_2^2 \end{aligned}$$

Therefore, the full per-timestep KL term becomes:

$$D_{\text{KL}} = \frac{\alpha_t}{2\kappa^2 \eta_{t-1} \eta_t} \|x_0 - f_\theta(x_t, y_0, t)\|_2^2$$

As discussed in Section 2, this coefficient is omitted in practice due to instability for small η_{t-1} .

B AI Usage

For this report, I declare that Gemini was used to improve grammar, while Claude assisted with coding, specifically in generating more visually appealing plots, partially generating the test suite and README, and providing auto-completion features.